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75) Simplification of Context Free Grammar

Reduction of CFG

Steps:

- 1) Reduction of CFG
- 2) Removal of Unit Productions
- 3) Removal of Null Production

1) CFG are reduced in two phases

Phase 1: Derivation of an equivalent grammar 'G', from the CFG, G, such that each variable derives some terminal string

Derivation Procedure:

Step 1: Include all symbols w_1 , that derives some terminal and initialize $i = 1$

Step 2: Include symbols w_{i+1} , that derives w_i

Step 3: Increment i and repeat Step 2, until $w_{i+1} = w_i$

Step 4: Include all production rules that w_i in it

Phase 2:

Derivation of an equivalent grammar G from the CFG, G' such that each symbol appears in a sentential form

Derivation Procedure:

Step 1: Include the Start Symbol in γ_1 and initialize $i=1$

Step 2: Include all symbols γ_{i+1} , that can be derived from γ_i and include all production rules that have been applied

Step 3: Increment i and repeat Step 2, until $\gamma_{i+1} = \gamma_i$

Example:

P: $S \rightarrow AC \mid B$, $A \rightarrow a$, $C \rightarrow c \mid BC$, $E \rightarrow aA \mid e$

Phase 1:

$$T = \{a, c, e\}$$

$$W_1 = \{A, C, E\}$$

$$W_2 = \{A, C, E, S\}$$



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 $W_3 = \{A, C, E, S\} \rightarrow \text{stop}$ $G' = \{(A, C, E, S), \{a, c, e\}, P, S\}$ $P: \begin{cases} S \rightarrow AC \\ A \rightarrow a \\ C \rightarrow c \\ E \rightarrow aA|e \end{cases}$

↑
can produce the
Terminal symbol

Phase 2:

 $\gamma_1 = \{S\}$ $\gamma_2 = \{A, C, S\}$ $\gamma_3 = \{S, A, C, a, c\}$ $\gamma_4 = \{S, A, C, a, c\} \rightarrow \text{stop}$ $G'' = \{(A, C, S), \{a, c\}, P, S\}$ $P: \underline{S \rightarrow AC, A \rightarrow a, C \rightarrow c}$

✓ can reach from the
Starting symbol

(76) Simplification of Context Free Grammar

Removal of Unit Production

Any Production Rule of the form $A \rightarrow B$ where $A, B \in \text{Non-Terminals}$ is called Unit Production

~~Produce~~

Procedure for Removal:

Step 1: To remove $A \rightarrow B$, add production $A \rightarrow x$ to the grammar rule whenever $B \rightarrow x$ occurs in the grammar. [$x \in \text{Terminal}$, x can be Null]

Step 2: Delete $A \rightarrow B$ from the grammar.

Step 3: Repeat from Step 1 until all Unit Productions are removed

Example:

P: $S \rightarrow XY$, $X \rightarrow a$, $Y \rightarrow z|b$, $z \rightarrow M$,
 $M \rightarrow N$, $N \rightarrow a$

$Y \rightarrow z$, $z \rightarrow M$, $M \rightarrow N$



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1) $M \rightarrow a$

Since $N \rightarrow a$, we can add $M \rightarrow a$

P: $S \rightarrow XY, X \rightarrow a, Y \rightarrow z|b, z \rightarrow M, M \rightarrow a, N \rightarrow a$

2) $z \rightarrow M$

Since $M \rightarrow a$, we can add $z \rightarrow a$

P: $S \rightarrow XY, X \rightarrow a, Y \rightarrow z|b, z \rightarrow a, M \rightarrow a, N \rightarrow a$

3) $Y \rightarrow z$

Since $z \rightarrow a$, we can add $Y \rightarrow a$

P: $S \rightarrow XY, X \rightarrow a, Y \rightarrow a|b, z \rightarrow a, M \rightarrow a, N \rightarrow a$

↓ Unreachable symbols

Phase 1: $W_1 = \{\$, X, Y, z, M, N\}$

$W_2 = \{S, X, Y, z, M, N\}$ all saved

Phase 2: $Y_1 = \{S\}$

$Y_2 = \{S, X, Y\}$

$Y_3 = \{S, X, Y, a, b\}$

So P: $\left\{ \begin{array}{l} S \rightarrow XY \\ X \rightarrow a \\ Y \rightarrow a|b \end{array} \right.$

⑦ Removal of Null Productions

In a CFG, a Non-Terminal Symbol 'A' is a nullable variable if there is a production $A \rightarrow \epsilon$ or there is a derivation that starts at 'A' and leads to ϵ . (Like $A \rightarrow \dots \rightarrow \epsilon$)

Procedure for Removal:

- Step 1: To remove $A \rightarrow \epsilon$, look for all productions whose right side contains A
- Step 2: Replace each occurrence of 'A' in each of these productions with ϵ
- Step 3: Add the resultant productions to the Grammar

Example:

$S \rightarrow ABAC, A \rightarrow aA | \epsilon, B \rightarrow bB | \epsilon, C \rightarrow C$

① $A \rightarrow \epsilon, B \rightarrow \epsilon$

1) to eliminate $A \rightarrow \epsilon$

$S \rightarrow ABAC$

$S \rightarrow ABC | BAC | \epsilon BC$

$A \rightarrow aA \quad A \rightarrow a$



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New Production:
$$S \rightarrow ABAC \mid ABC \mid BAC \mid BC$$
$$A \rightarrow aA \mid a, \quad B \rightarrow bB \mid \epsilon, \quad C \rightarrow c$$
2) To ~~off~~ eliminate $B \rightarrow \epsilon$
$$S \rightarrow AAC \mid AC \mid AC \mid C$$
$$B \rightarrow b$$
New Production:
$$S \rightarrow \cancel{AAB} ABAC \mid ABC \mid BAC \mid BC \mid \cancel{ABC} AC \mid C$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow bB \mid b$$
$$C \rightarrow c$$

(28)

Chomsky Normal FormCNF: we have a restriction ~~on the~~ :element in RHS should either be two variabor a Terminal